CTRM Add-In

A brief overview of reports and bookings

This document introduces the main functionalities of the CTRM Add-In. In particular, selected reports as well as ways to book transactions and update prices are introduced.

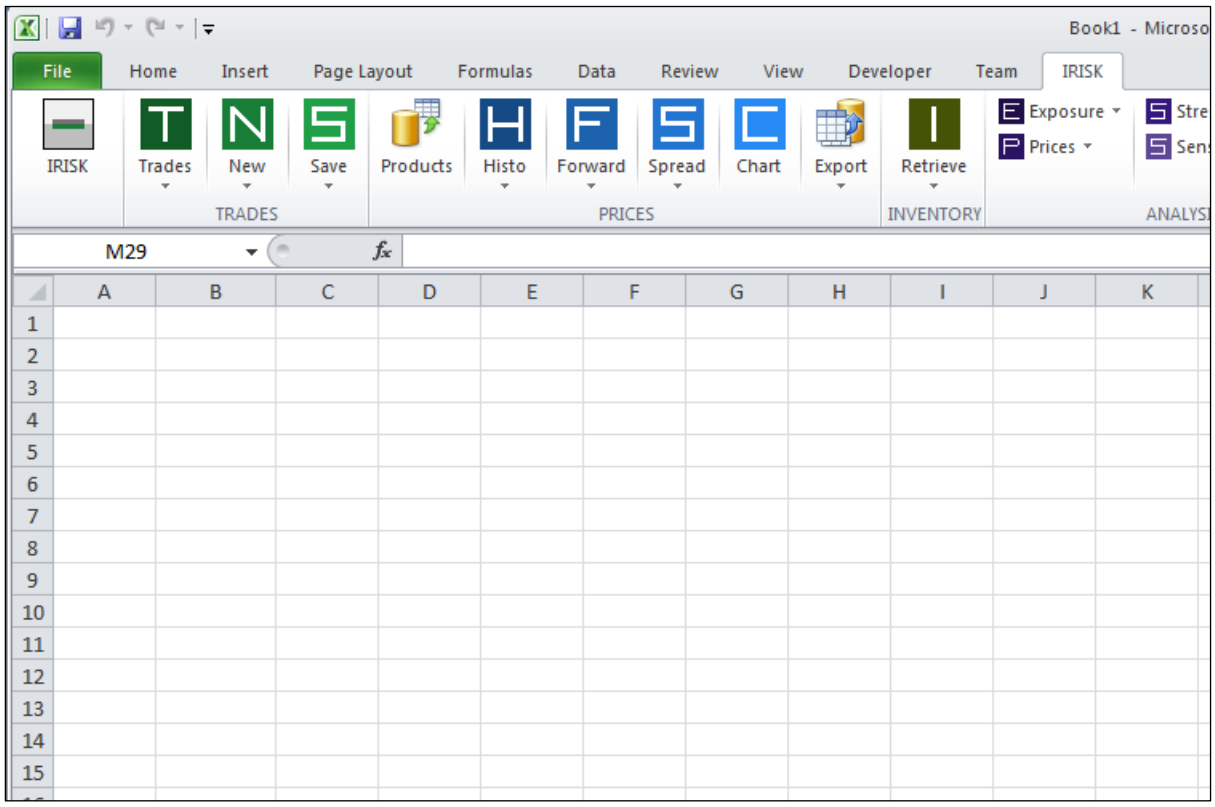
After a brief first section presenting the interface, selected reports are shown in the second section. Focus is on P&L, exposures and transactions. Third section highlights the input necessary to run the CTRM Add-In and update trades' valuation and risk metrics.

Table of Contents

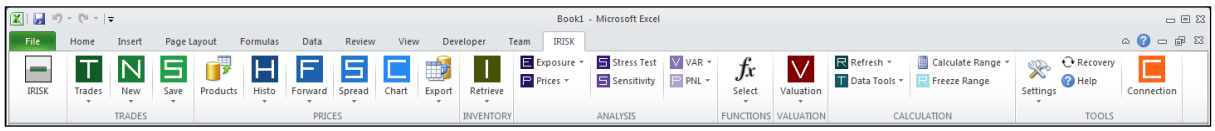
1. THE RIBBON 2
2. SELECTED REPORTS 4
   2.1.P&L 8
   2.2.Exposures 11
   2.3.Trades 13
   2.4.Other selected reports 14
   2.5.Ad-hoc reports using Excel-type functions 18
3. BOOKINGS 20
   3.1.Setup data 20
   3.2.Transactions 20
   3.3.Prices 23

1. THE RIBBON

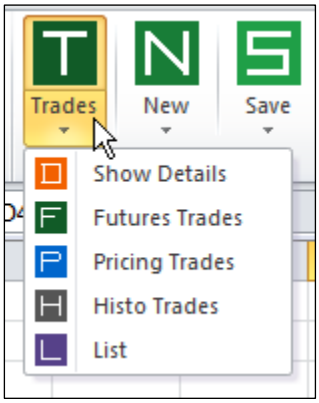
The CTRM Add-In's interface is fully integrated in MS Excel's ribbon.



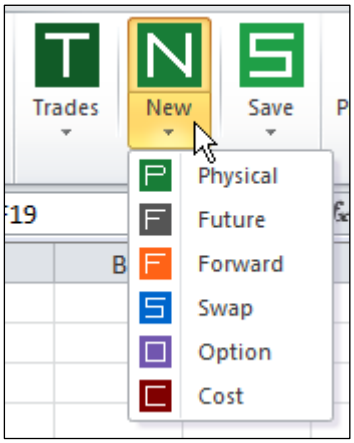
The CTRM Add-In's ribbon gives access to the system's different functionalities, including transaction booking, reporting and analysis.



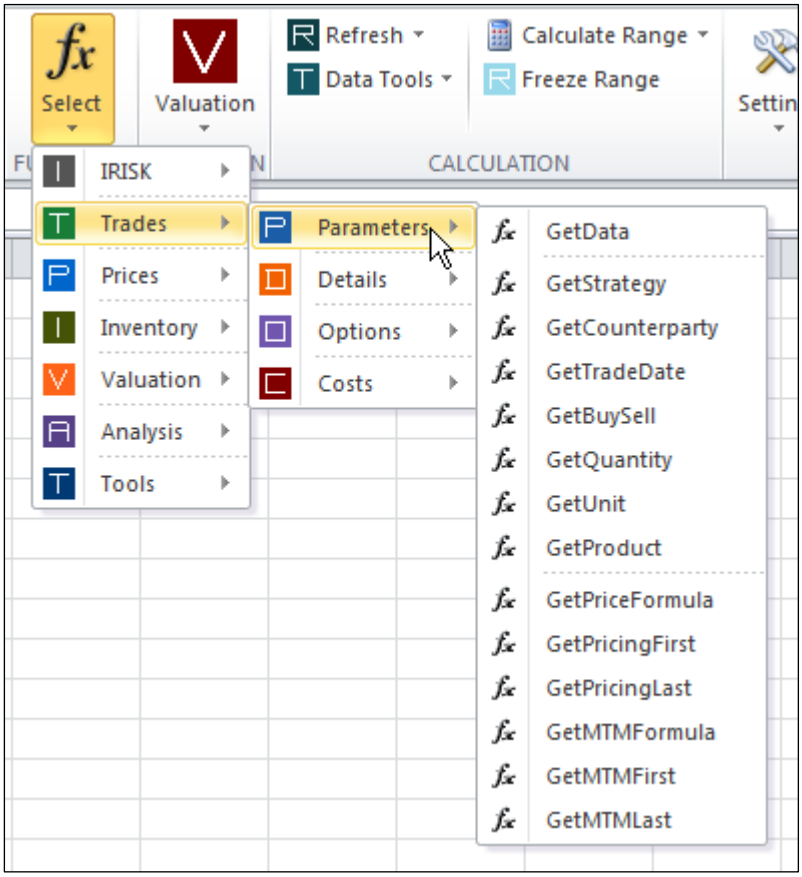
*As an illustration, transaction-related data and reports can be retrieved thanks to the **Trades** button and sub-buttons…*



*… New trades can be booked with the **New** buttons…*



… And more than 100 Excel-type functions can be used to retrieve trade data, assess transactions and perform analysis.



2. SELECTED REPORTS

Once trades are booked and prices updated *(see section 3. BOOKINGS)*, reports can be retrieved in several ways:

- **Built-in reports**
  Native reports give insight into P&L, exposures and other pricing & risk information. They can be retrieved in any Excel spreadsheet, at any chosen location, and give an overview of your trading metrics. A keyword specifies reports' locations within your spreadsheets, and a click on related buttons displays them.

- **Ad-hoc reports via Excel-type functions**
  Excel-type functions allow you to retrieve almost all booked trade information in any location you chose in your spreadsheets.

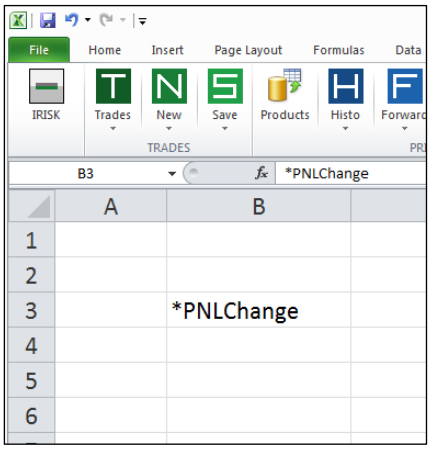
- **Ad-hoc reports via Excel VBA**
  Combining the two previously mentioned functionalities, you can create ad-hoc reports to view your trade data in the way you prefer. We can help you in setting up specific dashboards.

In more detail…

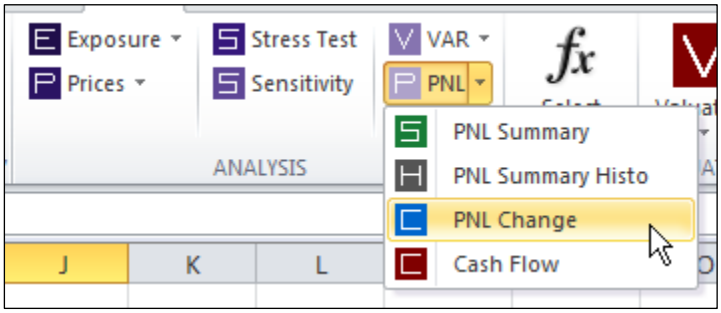
Built-in reports

Firstly, choose a report's destination within your spreadsheet. Secondly, enter the reports' keyword. Finally, click on the related button in the ribbon to display the report. For instance:

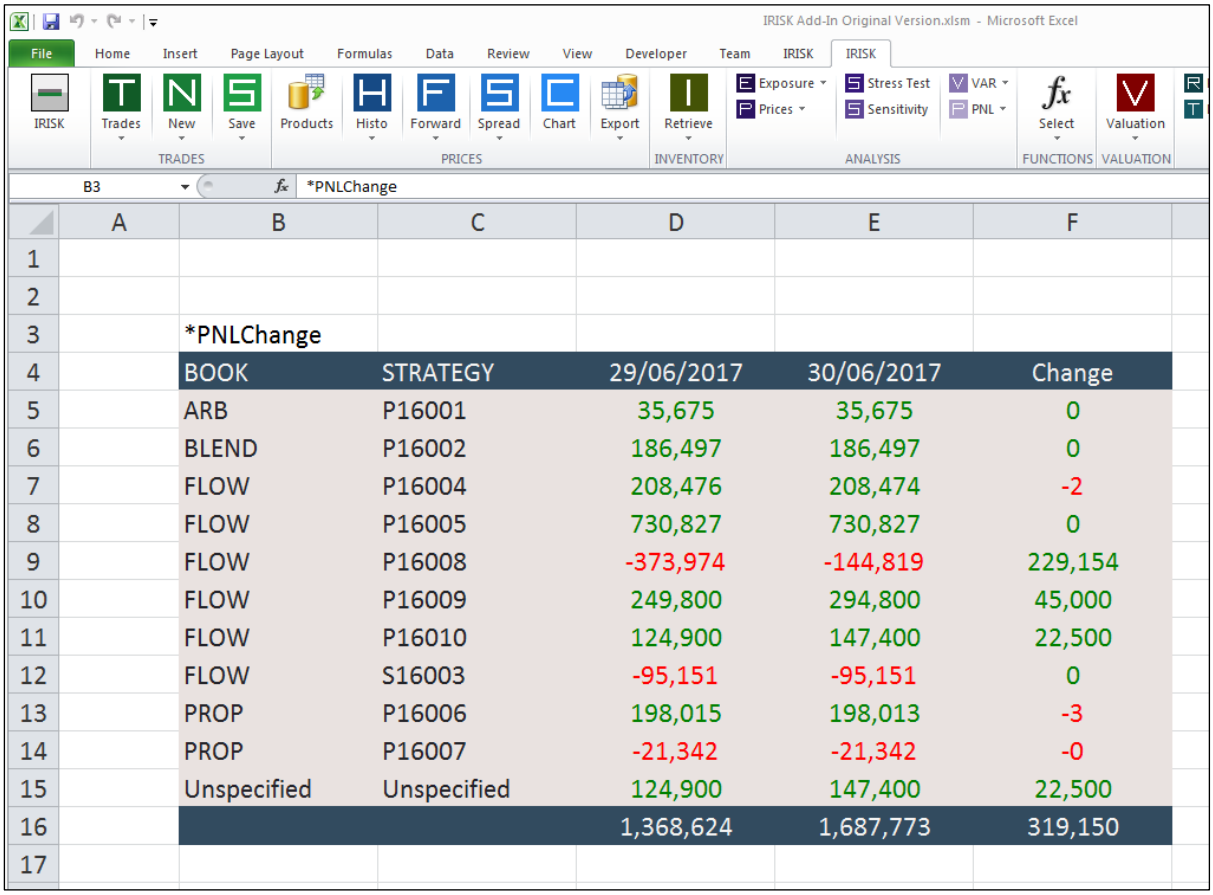
1. Insert "***PNLChange**" into a chosen cell:



2. Call the **PNLChange** report:



3. The related report then appears:



	A	B	C	D	E	F
1						
2						
3		*PNLChange				
4		BOOK	STRATEGY	29/06/2017	30/06/2017	Change
5		ARB	P16001	35,675	35,675	0
6		BLEND	P16002	186,497	186,497	0
7		FLOW	P16004	208,476	208,474	-2
8		FLOW	P16005	730,827	730,827	0
9		FLOW	P16008	-373,974	-144,819	229,154
10		FLOW	P16009	249,800	294,800	45,000
11		FLOW	P16010	124,900	147,400	22,500
12		FLOW	S16003	-95,151	-95,151	0
13		PROP	P16006	198,015	198,013	-3
14		PROP	P16007	-21,342	-21,342	-0
15		Unspecified	Unspecified	124,900	147,400	22,500
16				1,368,624	1,687,773	319,150
17						

NB: most reports can be parameterized. "PNL Change", for instance, could aggregate results into "books" or "strategies" only.

Ad-hoc reports via Excel-type functions

More than 100 functions notably allow retrieving trades' data, calculating P&L or assessing risk. For instance, the "**GetPNL()**" function retrieves P&L. It accepts the following fields as parameters: transaction ID, strategy, counterparty, bank, location, book or trader.

I19 | =GetPNL("STRATEGY","P16001")

H	I	J
	35,675	=GetPNL("STRATEGY","P16001")
PHYSICALS	40,000	=GetPNL("STRATEGY","P16001","PHYSICALS")
FUTURES	-2,325	=GetPNL("STRATEGY","P16001","FUTURES")
COSTS	-2,000	=GetPNL("STRATEGY","P16001","costs")

Combining built-in reports (***List()** below which lists all booked counterparties) with CTRM functions (**GetPNL()** below) allows to create new reports quickly:

L30 | =GetPNL("COUNTERPARTY",K30)

K	L	M
*LIST(COUNTERPARTY)		
COUNTERPARTY	**P&L**	
Bnk GBR Plc	-820,587	=GetPNL("COUNTERPARTY",K30)
Broker Bio SARL	-14,000	=GetPNL("COUNTERPARTY",K31)
Broker Fuel BV	-1,250	=GetPNL("COUNTERPARTY",K32)
Broker Oil Ltd	-2,500	=GetPNL("COUNTERPARTY",K33)
CHE Oil SARL	66,199	=GetPNL("COUNTERPARTY",K34)
CME	-2,300,298	=GetPNL("COUNTERPARTY",K35)
DEU Bio GmbH	2,607,565	=GetPNL("COUNTERPARTY",K36)
DEU Fuels GmbH	205,301	=GetPNL("COUNTERPARTY",K37)
FRA Bio SARL	58,455	=GetPNL("COUNTERPARTY",K38)
GBR Oil Ltd	-93,901	=GetPNL("COUNTERPARTY",K39)
ICE	528,050	=GetPNL("COUNTERPARTY",K40)
Inspector Bhd	-5,510	=GetPNL("COUNTERPARTY",K41)
Insure Oil GmbH	-8,750	=GetPNL("COUNTERPARTY",K42)
ITA Olio Srl	70,112	=GetPNL("COUNTERPARTY",K43)
MDEX	1,226,418	=GetPNL("COUNTERPARTY",K44)
MYS Bio Sdn Bhd	1,415,925	=GetPNL("COUNTERPARTY",K45)
NLD Bio BV	-18,455	=GetPNL("COUNTERPARTY",K46)
Oil Storage BV	-120,000	=GetPNL("COUNTERPARTY",K47)
Oil Tank Srl	-25,000	=GetPNL("COUNTERPARTY",K48)
Shipping Ltd	-1,080,000	=GetPNL("COUNTERPARTY",K49)

Ad-hoc reports via Excel VBA

Excel VBA can help in building ad-hoc reports, refreshing them, or executing otherwise too manual tasks. Purely illustrative, the below code reprices a dummy portfolio over the 10 last business days:

```
'********************
' 5 - Historical PNL
'********************
'
' This macro retrieves the ten latest P&L values for the full portfolio.
'
' - ValuationMessagesOn/ValuationMessagesOff turns the display of warning
'   messages on/off
' - GetLastDate(i) retrieves the ith latest date
' - GetHistoPNL("all") retrieves P&L for all the trades in the portfolio
'
' You can play with this code and change, for instance:
' - The days retrieved (for i = 1 to 100 for the latest 100 days)
' - The scope of the P&L specified by the variables in GetHistoPNL()
' - The location of the result in your spreadsheet (Cells(i,j) refers to
'   the cell with row i and column j)

Sub Demo_5_HistoricalPNL()
    Dim i As Long
    Dim dat As Date
    Dim changedate As Variant

    Application.Run "ValuationMessagesOff"
    For i = 1 To 10

        dat = Application.Run("GetLastDate", i)

        changedate = Application.Run("SetHistoDate", format(dat, "yyyymmdd"))

        Cells(2, 2) = "DATE"
        Cells(2, 3) = "P&L"

        Cells(i + 2, 2) = dat
        Cells(i + 2, 3) = Application.Run("GetHistoPNL", "all")
    Next i
    Application.Run "ValuationMessagesOn"

End Sub
```

It displays:

	A	B	C
1			
2		DATE	P&L
3		30/06/2017	1,687,773
4		29/06/2017	1,368,624
5		28/06/2017	1,247,753
6		27/06/2017	1,222,035
7		26/06/2017	1,271,967
8		23/06/2017	1,464,193
9		22/06/2017	1,386,708
10		21/06/2017	1,595,526
11		20/06/2017	1,683,492
12		19/06/2017	1,657,745
13			

The following sub-sections present a few selected reports.

2.1. P&L

As already shown, the CRTM Add-In grants several ways to look at P&L.

NB: P&L reporting currency is unique (e.g. USD), though different currencies can be used for transactions.

***PNLChange**

This displays P&L and its evolution over the last two business days.

*PNLChange

BOOK	STRATEGY	29/06/2017	30/06/2017	Change
ARB	P16001	35,675	35,675	0
BLEND	P16002	186,497	186,497	0
FLOW	P16004	208,476	208,474	-2
FLOW	P16005	730,827	730,827	0
FLOW	P16008	-373,974	-144,819	229,154
FLOW	P16009	249,800	294,800	45,000
FLOW	P16010	124,900	147,400	22,500
FLOW	S16003	-95,151	-95,151	0
PROP	P16006	198,015	198,013	-3
PROP	P16007	-21,342	-21,342	-0
Unspecified	Unspecified	124,900	147,400	22,500
		1,368,624	1,687,773	319,150

It can be parameterized:

*PNLChange(BOOK)

BOOK	29/06/2017	30/06/2017	Change
ARB	35,675	35,675	0
BLEND	186,497	186,497	0
FLOW	844,879	1,141,531	296,652
PROP	176,673	176,670	-3
Unspecified	124,900	147,400	22,500
	1,368,624	1,687,773	319,150

***PNLSummary**

This shows P&L for the last trading date only:

30/06/2017

=GetLastDate(1)

*PNLSummary

BOOK	STRATEGY	P&L
ARB	P16001	35,675
BLEND	P16002	186,497
FLOW	P16004	208,474
FLOW	P16005	730,827
FLOW	P16008	-144,819
FLOW	P16009	294,800
FLOW	P16010	147,400
FLOW	S16003	-95,151
PROP	P16006	198,013
PROP	P16007	-21,342
Unspecified	Unspecified	147,400
		1,687,773

NB: date has been added via a function.

***PNLSummaryHisto**

Historical date can be specified (and modified) in the Settings. PNLSummaryHisto displays P&L on that date.

30/12/2016

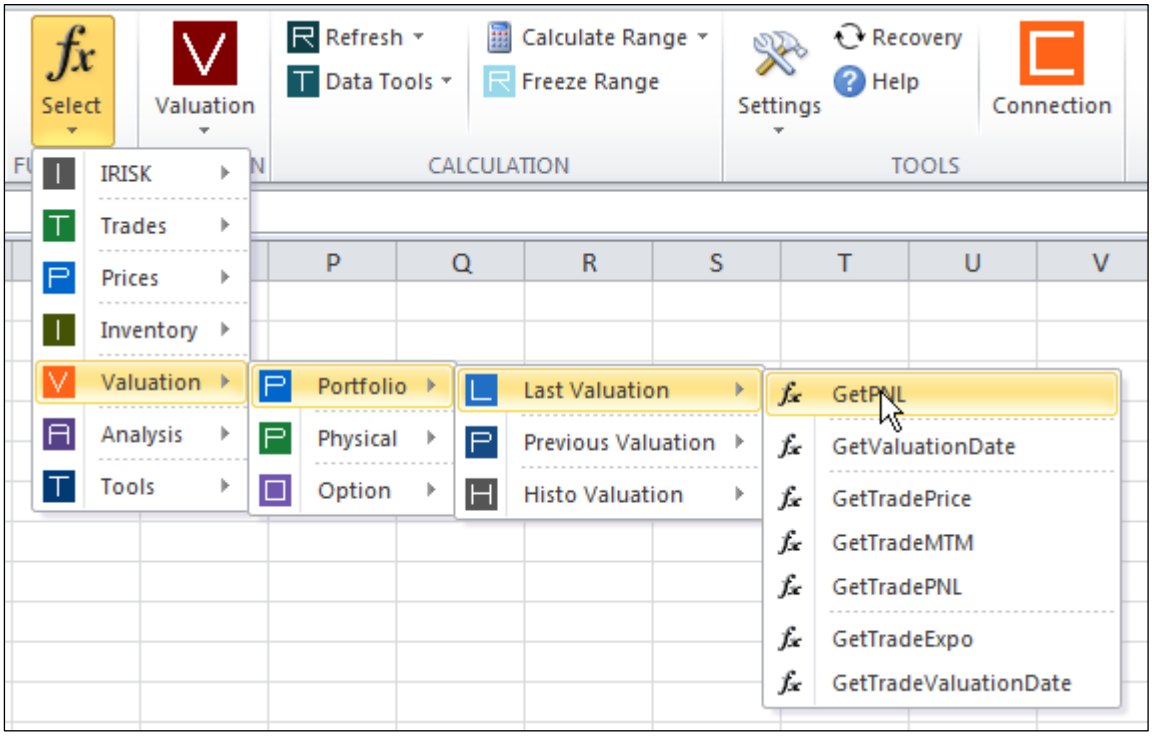
=GetHistoDate()

*PNLSummaryHisto

BOOK	STRATEGY	P&L
ARB	P16001	35,675
BLEND	P16002	186,497
FLOW	P16004	358,031
FLOW	P16005	1,636,448
FLOW	P16008	-730,629
FLOW	P16009	237,400
FLOW	S16003	-95,151
PROP	P16006	390,300
PROP	P16007	-21,328
		1,997,243

Other ways to report P&L include:

- The use of **Excel-type functions**:



- The use of **trades' report:**

	A	B	C	D	E	F	G	H	I
1									
2		*TRADES(ALL)							
3									
4		SUMMARY	QUANTITY	P&L					
5		PHYSICAL	-14,306	4,311,201					
6		FINANCIAL	-10,931	-1,366,417					
7		COSTS		-1,257,010					
8				1,687,773					
9									
10		*PHYSICALS							
11		ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT
12		PHY1000001	P16001	NLD Bio BV	05/05/2016	BUY	2,000	MT	FAME 0
13		PHY1000002	P16001	FRA Bio SARL	06/05/2016	SELL	-2,000	MT	FAME 0
14		PHY1000003	P16002	DEU Fuels GmbH	10/06/2016	BUY	2,500	MT	RME
15		PHY1000004	S16003	GBR Oil Ltd	22/06/2016	SELL	-1,250	MT	RME
16		PHY1000005	P16004	CHE Oil SARL	23/06/2016	BUY	14,000	bbl	FAME 0
17		PHY1000006	P16005	MYS Bio Sdn Bhd	29/06/2016	BUY	2,000	MT	PME
18		PHY1000007	P16005	MYS Bio Sdn Bhd	29/06/2016	BUY	2,000	MT	PME
19		PHY1000008	P16005	MYS Bio Sdn Bhd	29/06/2016	BUY	2,000	MT	PME

2.2. Exposures

Similarly to P&L, extracting exposures can be parameterized and done in several manners.

NB: if applicable, a unique reporting unit of measurement should be selected in order to aggregate exposures coherently.

***GetExpo()**

This retrieves overall positions:

*GetExpo(all)

30/06/2017	TOTAL	Jun17	Jul17	Aug17	Sep17	Oct17	Nov17	Dec17
ABI	-20,000		-2,000	-2,000	-2,000	-2,000	-2,000	-10,000
BO	16,112		4,028	4,028	4,028	4,028		
FAM	4,444	4,444						
G	6,000			2,000	-4,000	2,000	2,000	4,000
TOTAL	6,556	4,444	2,028	4,028	-1,972	4,028		-6,000

With specific parameters:

*GetExpo(strategy,P16008)

30/06/2017	TOTAL	Jun17	Jul17	Aug17	Sep17	Oct17	Nov17	Dec17
ABI	-12,000		-2,000	-2,000	-2,000	-2,000	-2,000	-2,000
BO	16,112		4,028	4,028	4,028	4,028		
G	-2,000			2,000	-4,000	2,000	2,000	-4,000
TOTAL	2,112		2,028	4,028	-1,972	4,028		-6,000

*GetExpo(book,flow)

30/06/2017	TOTAL	Jun17	Jul17	Aug17	Sep17	Oct17	Nov17	Dec17
ABI	-18,000		-2,000	-2,000	-2,000	-2,000	-2,000	-8,000
BO	16,112		4,028	4,028	4,028	4,028		
FAM	1,944	1,944						
G	4,000			2,000	-4,000	2,000	2,000	2,000
TOTAL	4,056	1,944	2,028	4,028	-1,972	4,028		-6,000

*GetExpo(book,flow,physicals)

30/06/2017	TOTAL	Jun17	Jul17	Aug17	Sep17	Oct17	Nov17	Dec17
ABI	-18,000		-2,000	-2,000	-2,000	-2,000	-2,000	-8,000
FAM	1,944	1,944						
G	16,000			2,000	2,000	2,000	2,000	8,000
TOTAL	-56	1,944	-2,000					

*EXPO()

Trade reports display exposure for each elementary transaction *(see 2.3. Trades)*. The command *EXPO() aggregates exposures selected as an Excel range (S12:S39 below):

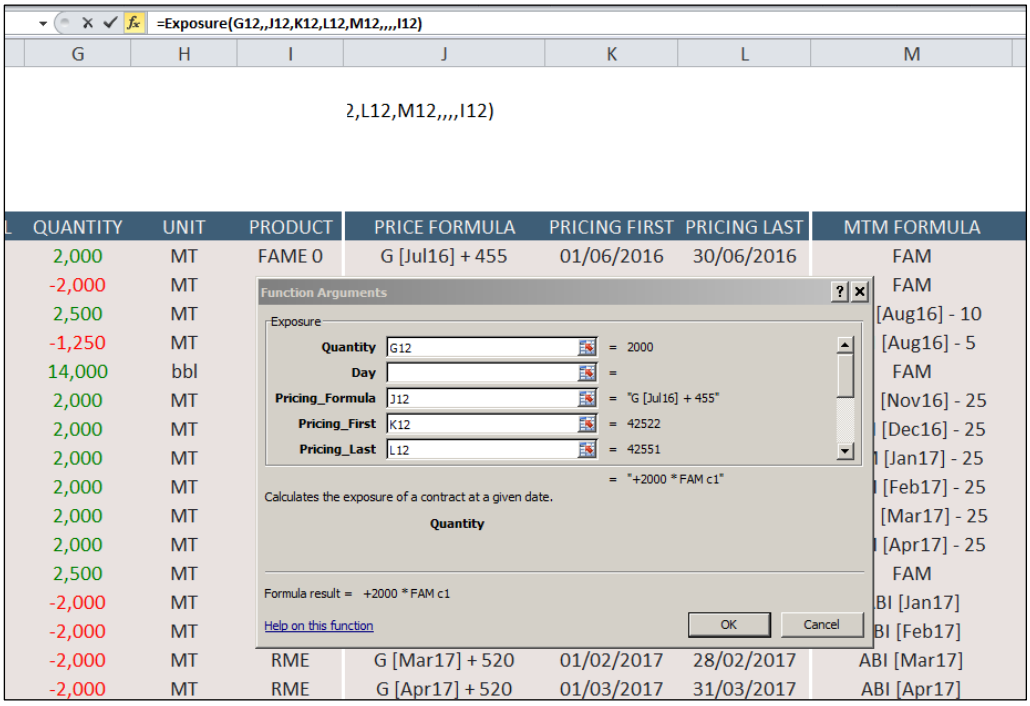
	R	S
10		
11	P&L	EXPOSURE
12	-18,455	+2000 * FAM c1
13	58,455	-2000 * FAM c1
14	205,301	
15	-93,901	
16	66,199	+1944.44 * FAM c1
17	334,304	
18	491,291	
19	104,811	
20	42,921	
21	234,341	
22	208,257	
23	70,112	+2500 * FAM c1
24	278,233	
25	284,962	
26	166,770	
27	206,618	
28	177,854	
29	154,921	
30	47,257	-2000 * ABI [Jul17]
31	118,608	-2000 * ABI [Aug17] +2000 * G [Aug17]
32	137,000	-2000 * ABI [Sep17] +2000 * G [Sep17]
33	149,364	-2000 * ABI [Oct17] +2000 * G [Oct17]
34	148,978	-2000 * ABI [Nov17] +2000 * G [Nov17]
35	147,400	-2000 * ABI [Dec17] +2000 * G [Dec17]
36	147,400	-2000 * ABI [Dec17] +2000 * G [Dec17]
37	147,400	-2000 * ABI [Dec17] +2000 * G [Dec17]
38	147,400	-2000 * ABI [Dec17] +2000 * G [Dec17]
39	147,400	-2000 * ABI [Dec17] +2000 * G [Dec17]

*EXPO(S12:S39)

30/06/2017	TOTAL	Jun17	Jul17	Aug17	Sep17	Oct17	Nov17	Dec17
ABI	-20,000		-2,000	-2,000	-2,000	-2,000	-2,000	-10,000
FAM	4,444	4,444						
G	18,000			2,000	2,000	2,000	2,000	10,000
TOTAL	2,444	4,444	-2,000					

=Exposure()

This Excel-type function calculates open exposure for a specific trade (physical or financial):



2.3. Trades

***TRADES()**

It retrieves all booking information related to a defined “transaction” (or all trades in portfolio).

*TRADES(STRATEGY,P16008)

SUMMARY	QUANTITY	P&L
PHYSICAL	-24,000	2,017,965
FINANCIAL	167	-2,143,284
COSTS		-19,500
		-144,819

*PHYSICALS

ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT	PRICE FORMULA	PRICING FIRST	PRICING LAST	MTM FORMULA	MTM FIRST	MTM LAST	PRICE	MTM	P&L	EXPOSURE
PHY1000033	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Jan17] + 520	01/12/2016	31/12/2016	ABI [Jan17]			1,003.81	864.69	278,233	
PHY1000034	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Feb17] + 520	01/01/2017	31/01/2017	ABI [Feb17]			1,008.38	865.90	284,962	
PHY1000035	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Mar17] + 520	01/02/2017	28/02/2017	ABI [Mar17]			1,015.95	932.57	166,770	
PHY1000036	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Apr17] + 520	01/03/2017	31/03/2017	ABI [Apr17]			986.09	882.78	206,618	
PHY1000037	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [May17] + 520	01/04/2017	30/04/2017	ABI [May17]			1,000.74	911.81	177,854	
PHY1000038	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Jun17] + 520	01/05/2017	31/05/2017	ABI [Jun17]			975.37	897.91	154,921	
PHY1000039	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Jul17] + 520	01/06/2017	30/06/2017	ABI [Jul17]			944.80	921.17	47,257	-2000 * ABI [Jul17]
PHY1000040	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Aug17] + 520	01/07/2017	31/07/2017	ABI [Aug17]			956.00	896.70	118,608	-2000 * ABI [Aug17] +2000 * G [Aug17]
PHY1000041	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Sep17] + 520	01/08/2017	31/08/2017	ABI [Sep17]			957.00	888.50	137,000	-2000 * ABI [Sep17] +2000 * G [Sep17]
PHY1000042	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Oct17] + 520	01/09/2017	30/09/2017	ABI [Oct17]			959.25	884.57	149,364	-2000 * ABI [Oct17] +2000 * G [Oct17]
PHY1000043	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Nov17] + 520	01/10/2017	31/10/2017	ABI [Nov17]			959.75	885.26	148,978	-2000 * ABI [Nov17] +2000 * G [Nov17]
PHY1000044	P16008	DEU Bio GmbH	28/11/2016	SELL	-2,000	MT	RME	G [Dec17] + 520	01/11/2017	30/11/2017	ABI [Dec17]			960.50	886.80	147,400	-2000 * ABI [Dec17] +2000 * G [Dec17]
					-24,000	MT										2,017,965	

*FUTURES

ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT	PRICE FORMULA	MTM FORMULA	PRICE	MTM	P&L	EXPOSURE
FUT1000096	P16008	ICE	28/11/2016	SELL	-60	lots	GO	438	G [Mar17]	438.00	465.00	-162,000	
FUT1000097	P16008	ICE	28/11/2016	SELL	-60	lots	GO	442	G [Jun17]	442.00	427.75	85,500	
FUT1000098	P16008	ICE	28/11/2016	SELL	-60	lots	GO	460	G [Sep17]	460.00	437.00	138,000	-6000 * G [Sep17]
FUT1000099	P16008	ICE	28/11/2016	SELL	-60	lots	GO	460.5	G [Dec17]	460.50	440.50	120,000	-6000 * G [Dec17]
FUT1000100	P16008	CME	28/11/2016	BUY	148	lots	BO	37.24	BO [Mar17]	37.24	31.85	-478,632	
FUT1000101	P16008	CME	28/11/2016	BUY	148	lots	BO	37.36	BO [May17]	37.36	32.63	-420,024	
FUT1000102	P16008	CME	28/11/2016	BUY	148	lots	BO	37.44	BO [Jul17]	37.44	32.93	-400,488	+4027.9 * BO [Jul17]
FUT1000103	P16008	CME	28/11/2016	BUY	148	lots	BO	37.31	BO [Aug17]	37.31	33.04	-379,176	+4027.9 * BO [Aug17]
FUT1000104	P16008	CME	28/11/2016	BUY	148	lots	BO	36.83	BO [Sep17]	36.83	33.16	-325,896	+4027.9 * BO [Sep17]
FUT1000105	P16008	CME	28/11/2016	BUY	148	lots	BO	36.78	BO [Oct17]	36.78	33.17	-320,568	+4027.9 * BO [Oct17]
					648	lots						-2,143,284	

*COSTS

ID	STRATEGY	COUNTERPARTY	TRADE DATE	COST TYPE	AMOUNT	UNIT	PRICE	MTM	P&L	EXPOSURE
COS1000016	P16008	Broker Bio SARL	28/11/2016	BROKER	-12,000	USD			-12,000	
COS1000017	P16008	Insure Oil GmbH	28/11/2016	INSURANCE	-7,500	USD			-7,500	
					-19,500	USD			-19,500	

Transaction’s **ID** is a unique identifier automatically generated. Except all four columns on the right (Price, MTM, P&L and Exposure), displayed data have been booked in the Add-In (see Section 3).

*TRADES() accepts any combination of the following fields as parameters:

Category	Trade Type
ID	ALL
STRATEGY	PHYSICALS
COUNTERPARTY	FUTURES
BANK	FORWARDS
LOCATION	SWAPS
BOOK	OPTIONS
TRADER	COSTS

For instance:

*TRADES(STRATEGY,P16008,FUTURES)

SUMMARY	QUANTITY	P&L
PHYSICAL		
FINANCIAL	167	-2,143,284
COSTS		
		-2,143,284

*FUTURES

ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT	PRICE FORMULA	MTM FORMULA	PRICE	MTM	P&L	EXPOSURE
FUT1000096	P16008	ICE	28/11/2016	SELL	-60	lots	GO	438	G [Mar17]	438.00	465.00	-162,000	
FUT1000097	P16008	ICE	28/11/2016	SELL	-60	lots	GO	442	G [Jun17]	442.00	427.75	85,500	
FUT1000098	P16008	ICE	28/11/2016	SELL	-60	lots	GO	460	G [Sep17]	460.00	437.00	138,000	-6000 * G [Sep17]
FUT1000099	P16008	ICE	28/11/2016	SELL	-60	lots	GO	460.5	G [Dec17]	460.50	440.50	120,000	-6000 * G [Dec17]
FUT1000100	P16008	CME	28/11/2016	BUY	148	lots	BO	37.24	BO [Mar17]	37.24	31.85	-478,632	
FUT1000101	P16008	CME	28/11/2016	BUY	148	lots	BO	37.36	BO [May17]	37.36	32.63	-420,024	
FUT1000102	P16008	CME	28/11/2016	BUY	148	lots	BO	37.44	BO [Jul17]	37.44	32.93	-400,488	+4027.9 * BO [Jul17]
FUT1000103	P16008	CME	28/11/2016	BUY	148	lots	BO	37.31	BO [Aug17]	37.31	33.04	-379,176	+4027.9 * BO [Aug17]
FUT1000104	P16008	CME	28/11/2016	BUY	148	lots	BO	36.83	BO [Sep17]	36.83	33.16	-325,896	+4027.9 * BO [Sep17]
FUT1000105	P16008	CME	28/11/2016	BUY	148	lots	BO	36.78	BO [Oct17]	36.78	33.17	-320,568	+4027.9 * BO [Oct17]
					648	lots						-2,143,284	

Optional “details” appear when clicking Trades > Show Details:

MTM FORMULA	BANK	LOCATION	B/L DATE	BOOK	TRADER	COMMENT	PRICE
G [Mar17]				FLOW	Claire Voyant		438.00
G [Jun17]				FLOW	Claire Voyant		442.00
G [Sep17]				FLOW	Claire Voyant		460.00
G [Dec17]				FLOW	Claire Voyant		460.50
BO [Mar17]				FLOW	Claire Voyant		37.24
BO [May17]				FLOW	Claire Voyant		37.36
BO [Jul17]				FLOW	Claire Voyant		37.44
BO [Aug17]				FLOW	Claire Voyant		37.31
BO [Sep17]				FLOW	Claire Voyant		36.83
BO [Oct17]				FLOW	Claire Voyant		36.78

*NB: *TRADES() report can serve to book new trades on the fly (another direct approach exists).*

2.4. Other selected reports

Further usual reports are introduced below.

***FuturesExpiry**

It shows open futures positions.

*FuturesExpiry

UNDERLYING	CONTRACT	LOTS	EXPIRY DATE
BO	BO [Jul17]	148	14/07/2017
BO	BO [Aug17]	148	14/08/2017
BO	BO [Sep17]	148	14/09/2017
BO	BO [Oct17]	148	13/10/2017
G	G [Sep17]	-60	12/09/2017
G	G [Dec17]	-60	12/12/2017

**FuturesTrades can also be used for a similar result*

*Cash Flow

It summarises cash flows expected from physical transactions.

*CashFlow

DATE	CF+	CF-	CASH FLOW
07/08/2017	1,912,000		1,912,000
07/09/2017	1,914,000		1,914,000
06/10/2017	1,918,500		1,918,500
07/11/2017	1,919,500		1,919,500
07/12/2017	9,605,000		9,605,000
	17,269,000	0	17,269,000

DATE	ID	STRATEGY	COUNTERPARTY	BUY/SELL	QUANTITY	UOM	PRODUCT	PRICING FORMULA	B/L DATE	PRICING LAST	PRICE	VALUE
07/08/2017	PHY1000040	P16008	DEU Bio GmbH	SELL	-2,000	MT	RME	G [Aug17] + 520		31/07/2017	956.00	1,912,000
07/09/2017	PHY1000041	P16008	DEU Bio GmbH	SELL	-2,000	MT	RME	G [Sep17] + 520		31/08/2017	957.00	1,914,000
06/10/2017	PHY1000042	P16008	DEU Bio GmbH	SELL	-2,000	MT	RME	G [Oct17] + 520		30/09/2017	959.25	1,918,500
07/11/2017	PHY1000043	P16008	DEU Bio GmbH	SELL	-2,000	MT	RME	G [Nov17] + 520		31/10/2017	959.75	1,919,500
07/12/2017	PHY1000044	P16008	DEU Bio GmbH	SELL	-2,000	MT	RME	G [Dec17] + 520		30/11/2017	960.50	1,921,000
07/12/2017	PHY1000045	P16009	DEU Bio GmbH	SELL	-2,000	MT	RME	G [Dec17] + 520		30/11/2017	960.50	1,921,000
07/12/2017	PHY1000046	P16009	DEU Bio GmbH	SELL	-2,000	MT	RME	G [Dec17] + 520		30/11/2017	960.50	1,921,000
07/12/2017	PHY1000047	P16010	DEU Bio GmbH	SELL	-2,000	MT	RME	G [Dec17] + 520		30/11/2017	960.50	1,921,000
07/12/2017	PHY1000048	Unspecified	DEU Bio GmbH	SELL	-2,000	MT	RME	G [Dec17] + 520		30/11/2017	960.50	1,921,000
												17,269,000

*PricingTrades(Month_Reference)

It displays contracts pricing in a given month (below all dummy contracts pricing in Oct16).

*PricingTrades(-28)

*PHYSICALS

ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT	PRICE FORMULA	PRICING FIRST	PRICING LAST	MTM FORMULA	MTM FIRST	MTM LAST	PRICE	MTM	P&L
PHY1000007	P16005	MYS Bio Sdn Bhd	29/06/2016	BUY	2,000	MT	PME	FCPO [Nov16] + 210	01/10/2016	31/10/2016	FAM [Dec16] - 25			697.00	942.65	491,291
					2,000	MT										491,291

*FUTURES

ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT	PRICE FORMULA	MTM FORMULA	PRICE	MTM	P&L
FUT1000036	P16005	ICE	29/06/2016	SELL	-20	lots	GO	443.75	G [Oct16]	443.75	469.25	-51,000
FUT1000037	P16005	ICE	29/06/2016	SELL	-20	lots	GO	444.50	G [Oct16]	444.50	469.25	-49,500
FUT1000042	P16005	MDEX	29/06/2016	BUY	80	lots	CPO	567.00	FUPO [Oct16]	567.00	628.25	122,500
FUT1000043	P16005	MDEX	29/06/2016	BUY	80	lots	CPO	566.25	FUPO [Oct16]	566.25	628.25	124,000
FUT1000058	P16007	MDEX	08/06/2016	SELL	-4	lots	CPO MYR	2573.50 MYR	FCPO [Oct16]	599.53	145.83	45,371
FUT1000059	P16007	MDEX	08/06/2016	SELL	-8	lots	CPO MYR	2571.25 MYR	FCPO [Oct16]	599.01	145.83	90,636
FUT1000060	P16007	MDEX	08/06/2016	SELL	-6	lots	CPO MYR	2557.50 MYR	FCPO [Oct16]	595.81	145.83	67,497
					102	lots						349,504

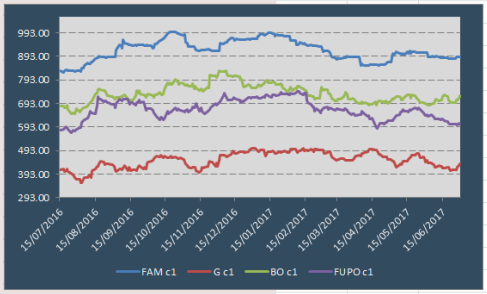
*SWAPS

ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT	PRICE FORMULA	PRICING FIRST	PRICING LAST	MTM FORMULA	MTM FIRST	MTM LAST	PRICE	MTM	P&L
SWA1000001	P16005	Bnk GBR Plc	30/06/2016	SELL	-2,000	MT	FAME/GO	G [Nov16] + 350	01/10/2016	31/10/2016	FAM [Nov16]	01/10/2016	31/10/2016	813.18	949.78	-273,212
					-2,000	MT										-273,212

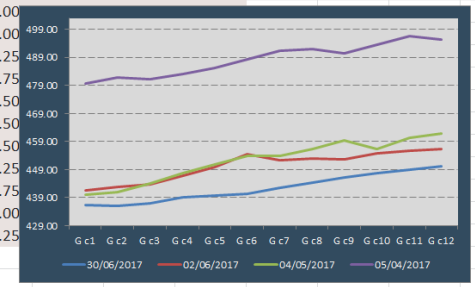
Historical Prices & Forward Curves

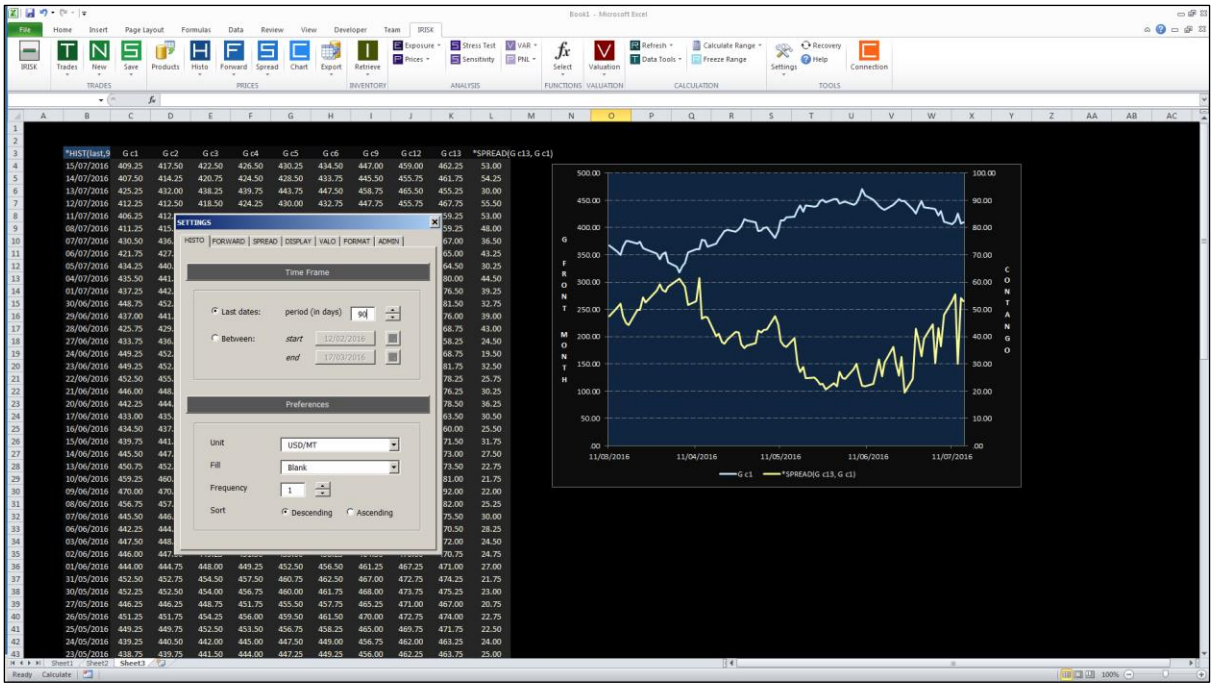
Historical prices, spreads as well as forward curves can be quickly retrieved.

*HIST(last,25	FAM c1	G c1	BO c1	FUPO c1
30/06/2017	891.55	436.50	725.98	604.50
29/06/2017	891.55	430.25	714.74	609.25
28/06/2017				
27/06/2017				
26/06/2017				
23/06/2017				
22/06/2017				
21/06/2017				
20/06/2017				
19/06/2017				
16/06/2017				
15/06/2017				
14/06/2017				
13/06/2017				
12/06/2017				
09/06/2017	892.94	420.50	711.87	628.25
08/06/2017	890.00	422.00	702.17	625.50
07/06/2017	890.36	432.50	691.37	635.00



*FWD(G)	30/06/2017	02/06/2017	04/05/2017	05/04/2017
G c1	436.50	441.50	440.00	479.75
G c2	436.00			
G c3	437.00			
G c4	439.25			
G c5	439.75			
G c6	440.50			
G c7	442.50			
G c8	444.50			
G c9	446.25			
G c10	447.75			
G c11	449.00			
G c12	450.25			





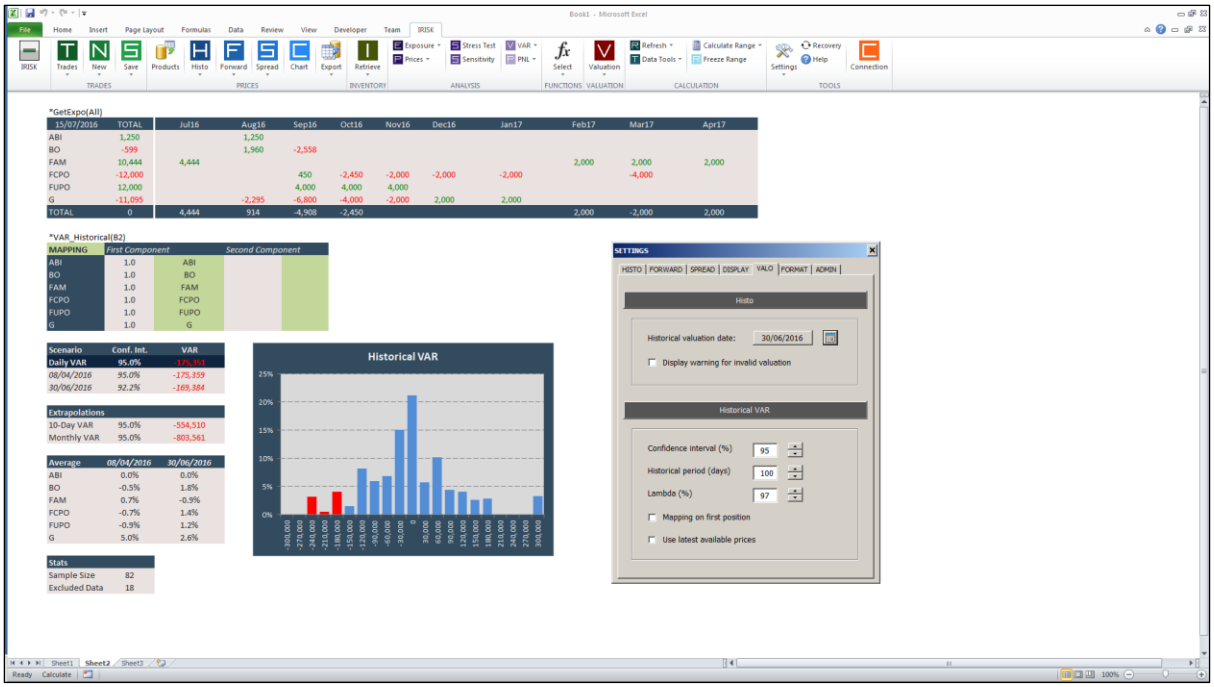
*Pricing()

It calculates and presents pricing formulas:

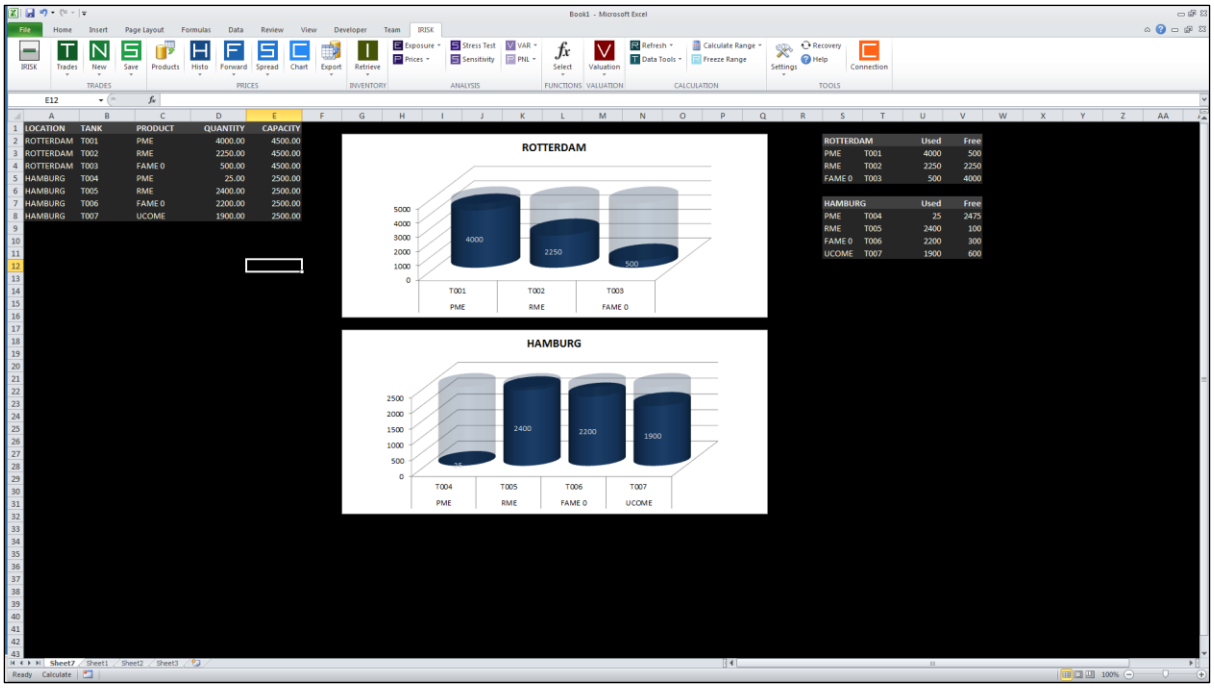
Buy	2000				
Price Formula	G [Jul17] + 415				
Pricing First	01/06/2017				
Pricing Last	30/06/2017				
MTM Formula	ABI c1				
Pricing Date	27/06/2017				
*PRICING(2000,27/06/2017,G [Jul17] + 415,01/06/2017,30/06/2017)					
Pricing Formula	G [Jul17] + 415	Pricing First	01/06/2017		
Quantity	2000	Pricing Last	30/06/2017		
Date	**27/06/2017**				
Priced Days	19	Priced Quantity	1727		
Unpriced Days	3	Unpriced Quantity	273		
Price	**837.51**	**Price in Base Unit**	**837.51**		
Pricing Days	Priced Days	Priced Quantity	Underlying	Premium	Price
01/06/2017	1	91	451.25	415.00	866.25
02/06/2017	2	182	442.75	415.00	857.75
05/06/2017	3	273	439.75	415.00	854.75
06/06/2017	4	364	434.50	415.00	849.50
07/06/2017	5	455	433.25	415.00	848.25
08/06/2017	6	545	422.75	415.00	837.75
09/06/2017	7	636	422.00	415.00	837.00
12/06/2017	8	727	426.00	415.00	841.00
13/06/2017	9	818	424.75	415.00	839.75
14/06/2017	10	909	427.25	415.00	842.25
15/06/2017	11	1000	418.75	415.00	833.75
16/06/2017	12	1091	419.25	415.00	834.25
19/06/2017	13	1182	422.75	415.00	837.75
20/06/2017	14	1273	420.75	415.00	835.75
21/06/2017	15	1364	413.25	415.00	828.25
22/06/2017	16	1455	406.25	415.00	821.25
23/06/2017	17	1545	411.25	415.00	826.25
26/06/2017	18	1636	409.75	415.00	824.75
27/06/2017	19	1727	412.25	415.00	827.25
28/06/2017			412.25	415.00	827.25
29/06/2017			412.25	415.00	827.25
30/06/2017			412.25	415.00	827.25
	19/22	1727	422.51	415.00	837.51

Value At Risk

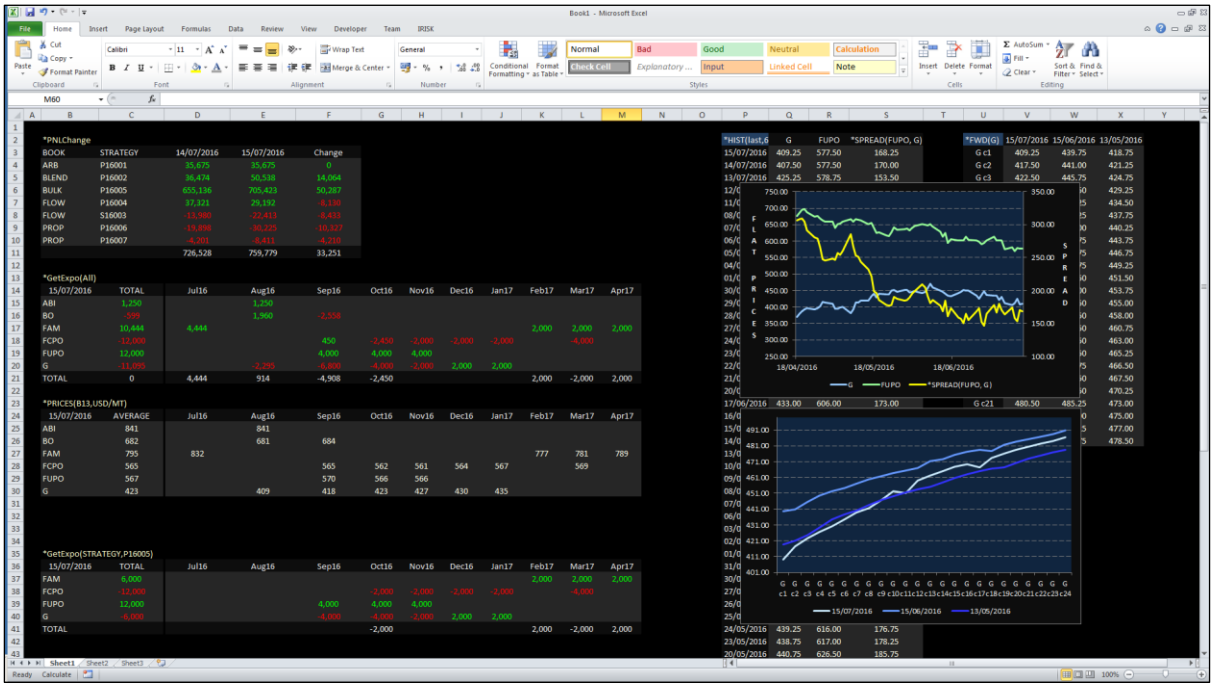
VAR calculation, Stress Testing as well as Potential Future Exposure (credit risk measure) can be performed:



Inventory report

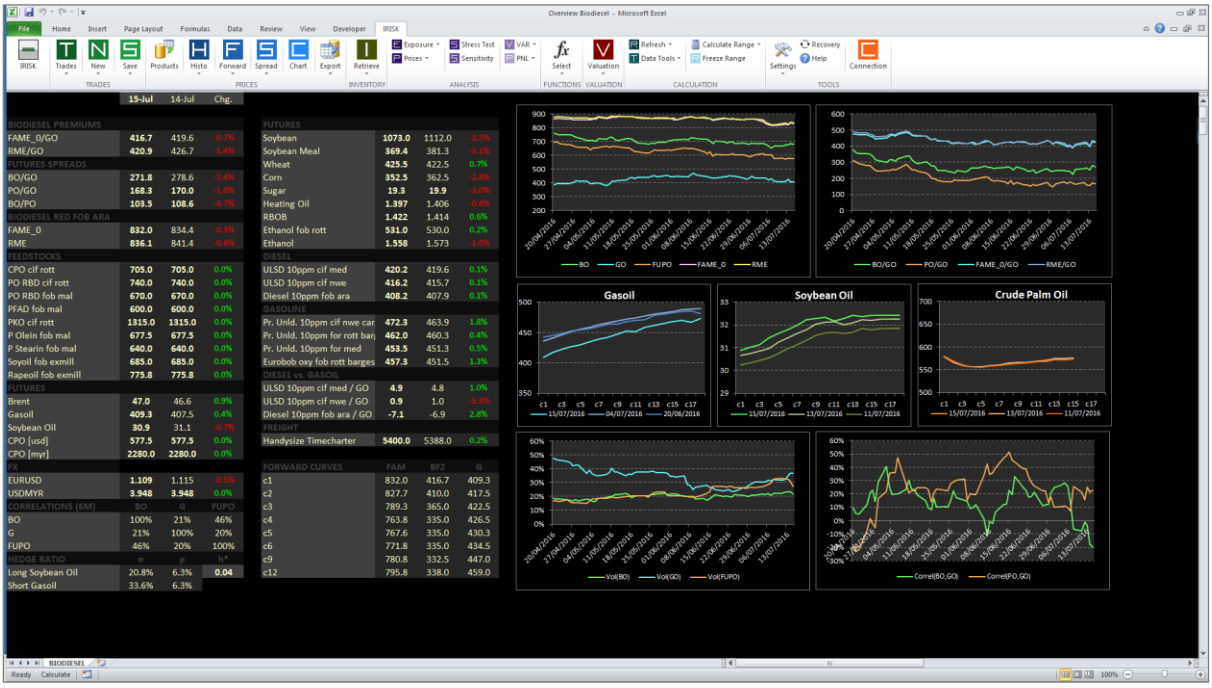


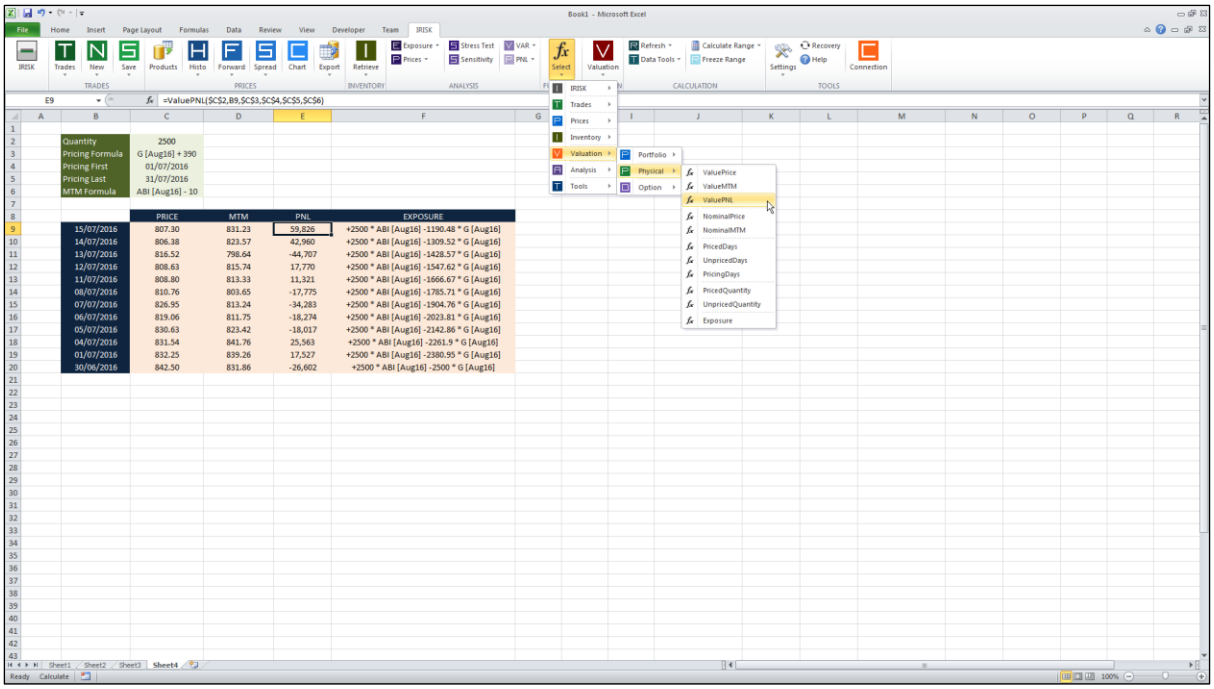
Combination of reports



2.5. Ad-hoc reports using Excel-type functions

Dashboards can be created thanks to Excel-type functions. Some examples are shown below.




Book1 - Microsoft Excel
E9
=ValuePNL(C2,B9,C3,C4,C5,C6)
Quantity 2500
Pricing Formula G [Aug16] + 390
Pricing First 01/07/2016
Pricing Last 31/07/2016
MTM Formula ABI [Aug16] - 10
PRICE MTM PNL EXPOSURE
15/07/2016 807.30 831.23 59,826 +2500 * ABI [Aug16] -1190.48 * G [Aug16]
14/07/2016 806.38 823.57 42,960 +2500 * ABI [Aug16] -1309.52 * G [Aug16]
13/07/2016 816.52 798.64 -44,707 +2500 * ABI [Aug16] -1428.57 * G [Aug16]
12/07/2016 808.63 815.74 17,770 +2500 * ABI [Aug16] -1547.62 * G [Aug16]
11/07/2016 808.80 813.33 11,321 +2500 * ABI [Aug16] -1666.67 * G [Aug16]
08/07/2016 810.76 803.65 -17,775 +2500 * ABI [Aug16] -1785.71 * G [Aug16]
07/07/2016 826.95 813.24 -34,283 +2500 * ABI [Aug16] -1904.76 * G [Aug16]
06/07/2016 819.06 811.75 -18,274 +2500 * ABI [Aug16] -2023.81 * G [Aug16]
05/07/2016 830.63 823.42 -18,017 +2500 * ABI [Aug16] -2142.86 * G [Aug16]
04/07/2016 831.54 841.76 25,563 +2500 * ABI [Aug16] -2261.9 * G [Aug16]
01/07/2016 832.25 839.26 17,527 +2500 * ABI [Aug16] -2380.95 * G [Aug16]
30/06/2016 842.50 831.86 -26,602 +2500 * ABI [Aug16] -2500 * G [Aug16]
IRISK
Trades
Prices
Inventory
Valuation
Analysis
Tools
Portfolio
Physical
Option
ValuePrice
ValueMTM
ValuePNL
NominalPrice
NominalMTM
PricedDays
UnpricedDays
PricingDays
PricedQuantity
UnpricedQuantity
Exposure

3. BOOKINGS

The following datasets are required in order to operate the CTRM Add-In smoothly:

- Setup data
- Transactions
- Prices

3.1. Setup data

At inception, the followings must be initialized. We will perform this for you.

- **Reporting currency**
  Unique, all P&L will be displayed in this unit (for instance USD).

- **Exposure's Unit of Measurement**
  If applicable, a common exposure unit shall be chosen in order to allow exposures aggregation (e.g. MT, bbl…).

- **Lot sizes and conversion factors**
  Lot sizes and conversion between units (MT, bbl, bu, lbs…), when applicable, shall be defined.

- **Calendars**
  Holidays and expiration calendars shall be booked.

3.2. Transactions

The tables below gather required booking data for physical and financial transactions.

NECESSARY	RME
OPTIONAL	DEU Bio GmbH

*PHYSICALS	
ID	PHY1000033
STRATEGY	P16008
COUNTERPARTY	DEU Bio GmbH
TRADE DATE	28/11/2016
BUY/SELL	SELL
QUANTITY	-2,000
UNIT	MT
PRODUCT	RME
PRICE FORMULA	G [Jan17] + 520
PRICING FIRST	01/12/2016
PRICING LAST	31/12/2016
MTM FORMULA	ABI [Jan17]
MTM FIRST	
MTM LAST	
BANK	
LOCATION	
B/L DATE	
BOOK	FLOW
TRADER	Claire Voyant
COMMENT	

*FUTURES *FORWARDS *SWAPS	
ID	FUT1000096
STRATEGY	P16008
COUNTERPARTY	ICE
TRADE DATE	28/11/2016
BUY/SELL	SELL
QUANTITY	-60
UNIT	lots
PRODUCT	GO
PRICE FORMULA	438
PRICING FIRST*	
PRICING LAST*	
MTM FORMULA	G [Mar17]
MTM FIRST*	
MTM LAST*	
BANK	
LOCATION	
B/L DATE	
BOOK	FLOW
TRADER	Claire Voyant
COMMENT	

** not required for futures*

*COSTS	
ID	COS1000016
STRATEGY	P16008
COUNTERPARTY	Broker Bio SARL
TRADE DATE	28/11/2016
COST TYPE	BROKER
AMOUNT	-12,000
UNIT	USD
BANK	
LOCATION	
B/L DATE	
BOOK	FLOW
TRADER	Claire Voyant
COMMENT	

Apart from basic trade information (trade date, purchase/sale, quantity, unit or currency, product), the following pieces of information are required:

Price Formula: Transaction's price. It can either be:

Fixed

For a price not libelled in the CTRM's reporting currency, currency shall be specified (e.g. "**438**", "**438 USD**", "**2571.25 MYR**"…)

Floating

It can depend on a specific contract ("**G [Jan17]**" standing for ICE Gasoil Jan17 Future), a Spot price ("**FAM**" for FAME 0 Spot price), a continuous nearby contract ("**G c1**" for the 1st continuously rolling ICE Gasoil Future). Premium or discount can be added ("**G [Jan17] + 520**", "**FCPO [Oct16] + 210**").

Pricing First: First pricing day, if applicable

Pricing Last: Last pricing day, if applicable

MTM Formula: Similar to **Price Formula**

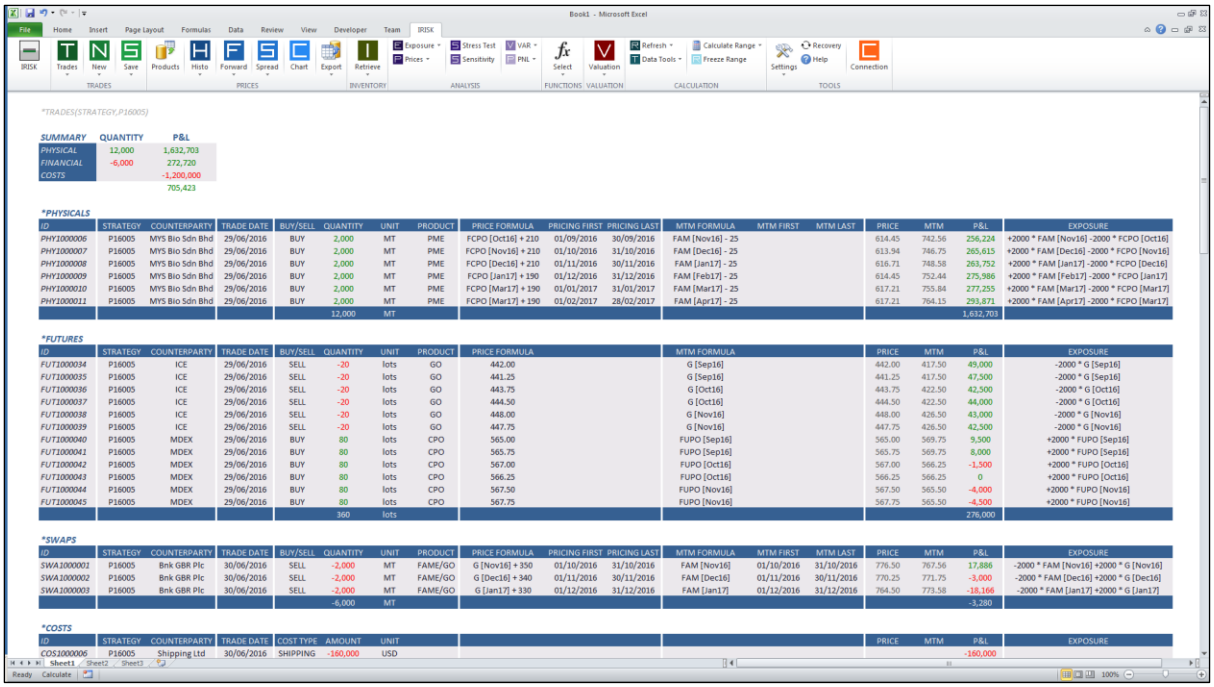
MTM First: First MTM day, if applicable

MTM Last: Last MTM day, if applicable

Some examples below *(price, MTM, P&L and exposures are calculated)*:

*FUTURES

ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT	PRICE FORMULA	MTM FORMULA	PRICE	MTM	P&L
FUT1000055	P16007	MDEX	07/06/2016	BUY	6	lots	CPO MYR	2572.25 MYR	FCPO [Sep16]	599.24	684.30	12,758
FUT1000056	P16007	MDEX	07/06/2016	BUY	5	lots	CPO MYR	2568.50 MYR	FCPO [Sep16]	598.37	684.30	10,741
FUT1000057	P16007	MDEX	07/06/2016	BUY	7	lots	CPO MYR	2570.00 MYR	FCPO [Sep16]	598.72	684.30	14,977
FUT1000058	P16007	MDEX	08/06/2016	SELL	-4	lots	CPO MYR	2573.50 MYR	FCPO [Oct16]	599.53	625.97	-2,643
FUT1000059	P16007	MDEX	08/06/2016	SELL	-8	lots	CPO MYR	2571.25 MYR	FCPO [Oct16]	599.01	625.97	-5,391
FUT1000060	P16007	MDEX	08/06/2016	SELL	-6	lots	CPO MYR	2557.50 MYR	FCPO [Oct16]	595.81	625.97	-4,524
FUT1000061	P16007	ICE	09/06/2016	BUY	2	lots	GO	469.75	G [Aug16]	469.75	377.50	-18,450
FUT1000062	P16007	ICE	09/06/2016	BUY	4	lots	GO	468.25	G [Aug16]	468.25	377.50	-36,300
FUT1000063	P16007	ICE	09/06/2016	BUY	1	lots	GO	470.50	G [Aug16]	470.50	377.50	-9,300



*TRADES(STRATEGY,P16005)

SUMMARY	QUANTITY	P&L
PHYSICAL	12,000	1,632,703
FINANCIAL	-6,000	272,720
COSTS		-1,200,000
		705,423

*PHYSICALS

ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT	PRICE FORMULA	PRICING FIRST	PRICING LAST	MTM FORMULA	MTM FIRST	MTM LAST	PRICE	MTM	P&L	EXPOSURE
PHY1000006	P16005	MYS Bio Sdn Bhd	29/06/2016	BUY	2,000	MT	PME	FCPO [Oct16] + 210	01/09/2016	30/09/2016	FAM [Nov16] - 25			614.45	742.56	256,224	+2000 * FAM [Nov16] -2000 * FCPO [Oct16]
PHY1000007	P16005	MYS Bio Sdn Bhd	29/06/2016	BUY	2,000	MT	PME	FCPO [Nov16] + 210	01/10/2016	31/10/2016	FAM [Dec16] - 25			613.94	746.75	265,615	+2000 * FAM [Dec16] -2000 * FCPO [Nov16]
PHY1000008	P16005	MYS Bio Sdn Bhd	29/06/2016	BUY	2,000	MT	PME	FCPO [Dec16] + 210	01/11/2016	30/11/2016	FAM [Jan17] - 25			616.71	748.58	263,752	+2000 * FAM [Jan17] -2000 * FCPO [Dec16]
PHY1000009	P16005	MYS Bio Sdn Bhd	29/06/2016	BUY	2,000	MT	PME	FCPO [Jan17] + 190	01/12/2016	31/12/2016	FAM [Feb17] - 25			614.45	752.44	275,986	+2000 * FAM [Feb17] -2000 * FCPO [Jan17]
PHY1000010	P16005	MYS Bio Sdn Bhd	29/06/2016	BUY	2,000	MT	PME	FCPO [Mar17] + 190	01/01/2017	31/01/2017	FAM [Mar17] - 25			617.21	755.84	277,255	+2000 * FAM [Mar17] -2000 * FCPO [Mar17]
PHY1000011	P16005	MYS Bio Sdn Bhd	29/06/2016	BUY	2,000	MT	PME	FCPO [Mar17] + 190	01/02/2017	28/02/2017	FAM [Apr17] - 25			617.21	764.15	293,871	+2000 * FAM [Apr17] -2000 * FCPO [Mar17]
					12,000	MT										1,632,703	

*FUTURES

ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT	PRICE FORMULA	MTM FORMULA	PRICE	MTM	P&L	EXPOSURE
FUT1000034	P16005	ICE	29/06/2016	SELL	-20	lots	GO	442.00	G [Sep16]	442.00	417.50	49,000	-2000 * G [Sep16]
FUT1000035	P16005	ICE	29/06/2016	SELL	-20	lots	GO	441.25	G [Sep16]	441.25	417.50	47,500	-2000 * G [Sep16]
FUT1000036	P16005	ICE	29/06/2016	SELL	-20	lots	GO	443.75	G [Oct16]	443.75	422.50	42,500	-2000 * G [Oct16]
FUT1000037	P16005	ICE	29/06/2016	SELL	-20	lots	GO	444.50	G [Oct16]	444.50	422.50	44,000	-2000 * G [Oct16]
FUT1000038	P16005	ICE	29/06/2016	SELL	-20	lots	GO	448.00	G [Nov16]	448.00	426.50	43,000	-2000 * G [Nov16]
FUT1000039	P16005	ICE	29/06/2016	SELL	-20	lots	GO	447.75	G [Nov16]	447.75	426.50	42,500	-2000 * G [Nov16]
FUT1000040	P16005	MDEX	29/06/2016	BUY	80	lots	CPO	565.00	FUPO [Sep16]	565.00	569.75	9,500	+2000 * FUPO [Sep16]
FUT1000041	P16005	MDEX	29/06/2016	BUY	80	lots	CPO	565.75	FUPO [Sep16]	565.75	569.75	8,000	+2000 * FUPO [Sep16]
FUT1000042	P16005	MDEX	29/06/2016	BUY	80	lots	CPO	567.00	FUPO [Oct16]	567.00	566.25	-1,500	+2000 * FUPO [Oct16]
FUT1000043	P16005	MDEX	29/06/2016	BUY	80	lots	CPO	566.25	FUPO [Oct16]	566.25	566.25	0	+2000 * FUPO [Oct16]
FUT1000044	P16005	MDEX	29/06/2016	BUY	80	lots	CPO	567.50	FUPO [Nov16]	567.50	565.50	-4,000	+2000 * FUPO [Nov16]
FUT1000045	P16005	MDEX	29/06/2016	BUY	80	lots	CPO	567.75	FUPO [Nov16]	567.75	565.50	-4,500	+2000 * FUPO [Nov16]
					360	lots						276,000	

*SWAPS

ID	STRATEGY	COUNTERPARTY	TRADE DATE	BUY/SELL	QUANTITY	UNIT	PRODUCT	PRICE FORMULA	PRICING FIRST	PRICING LAST	MTM FORMULA	MTM FIRST	MTM LAST	PRICE	MTM	P&L	EXPOSURE
SWA1000001	P16005	Bnk GBR Plc	30/06/2016	SELL	-2,000	MT	FAME/GO	G [Nov16] + 350	01/10/2016	31/10/2016	FAM [Nov16]	01/10/2016	31/10/2016	776.50	767.56	17,886	-2000 * FAM [Nov16] +2000 * G [Nov16]
SWA1000002	P16005	Bnk GBR Plc	30/06/2016	SELL	-2,000	MT	FAME/GO	G [Dec16] + 340	01/11/2016	30/11/2016	FAM [Dec16]	01/11/2016	30/11/2016	770.25	771.75	-3,000	-2000 * FAM [Dec16] +2000 * G [Dec16]
SWA1000003	P16005	Bnk GBR Plc	30/06/2016	SELL	-2,000	MT	FAME/GO	G [Jan17] + 330	01/12/2016	31/12/2016	FAM [Jan17]	01/12/2016	31/12/2016	764.50	773.58	-18,166	-2000 * FAM [Jan17] +2000 * G [Jan17]
					-6,000	MT										-3,280	

*COSTS

ID	STRATEGY	COUNTERPARTY	TRADE DATE	COST TYPE	AMOUNT	UNIT	PRICE	MTM	P&L	EXPOSURE
COS1000006	P16005	Shipping Ltd	30/06/2016	SHIPPING	-160,000	USD			-160,000	

3.3. Prices

On a regular basis (e.g. daily), commodity forward curves shall be booked in the following manner (template appears when clicking Export > Template Columns).

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	ENTER DATE	ABI	ACF	BFZ	BO	BRI	EURUSD	FAM	FCPO	FUPO	G	RSO	RSO_EUR	S
2	30/06/2017	Argus Biodi	Ethanol Fut	Argus Biodi	Soybean Oi	Argus Biodi	CME EURO I	Argus Biodi	Crude Palm	Crude Palm	G-Gasoil Fu	Rapeseed C	Rapeseed C	Soybean Fu
3	c1	897.91	1.51	467.01	32.93	473.38	1.14	891.55	2596.00	604.50	436.50	817.53	715.50	942.50
4	c2	921.17	1.52	476.00	33.04	485.00	1.14	912.17	2497.00	581.50	436.00	812.39	711.00	947.00
5	c3	896.70	1.52	436.00	33.16	460.00	1.15	872.70	2459.00	572.75	437.00	809.25	708.25	951.00
6	c4	888.50	1.51	420.00	33.17	450.00	1.15	858.50	2448.00	570.25	439.25	809.25	708.25	955.50
7	c5	884.57	1.50	381.00	33.34	445.00	1.15	820.57	2455.00	571.75	439.75	809.25	708.25	963.00
8	c6	885.26	1.49	381.00	33.50	445.00	1.16	821.26	2472.00	575.75	440.50	809.25	708.25	967.50
9	c7	886.80	1.47	381.00	33.59	445.00	1.16	822.80	2502.00	582.75	442.50	809.25	708.25	973.50
10	c8	878.86	1.47	365.00	33.66	435.00	1.17	808.86	2524.00	587.75	444.50	809.25	708.25	978.50
11	c9	880.64	1.47	365.00	33.74	435.00	1.18	810.64	2543.00	592.25	446.25	809.25	708.25	976.00
12	c10	882.25	1.50	365.00	33.72	435.00	1.18	812.25	2515.00	585.75	447.75	809.25	708.25	967.00
13	c11	873.52	1.50	365.00	33.67	425.00	1.19	813.52	2507.00	584.00	449.00			960.50
14	c12	874.87	1.49	365.00	33.40	425.00	1.20	814.87	2498.00	581.75	450.25			965.50
15														

Historical time series can also be inserted as follows.

	A	B	C	D	E
1	CODE	CONTRACT	DAY	PRICE	COMMENT
2	BO c1	[May16]	04/05/2016	32.20	
3	BO c1	[May16]	05/05/2016	32.01	
4	BO c1	[May16]	06/05/2016	32.81	
5	BO c1	[May16]	09/05/2016	32.54	
6	BO c1	[May16]	10/05/2016	33.13	
7	BO c1	[May16]	11/05/2016	33.08	
8	BO c1	[May16]	12/05/2016	32.30	
9	BO c1	[May16]	13/05/2016	32.17	
10	BO c1	[Jul16]	16/05/2016	32.79	
11	BO c1	[Jul16]	17/05/2016	32.78	
12	BO c1	[Jul16]	18/05/2016	32.45	
13	BO c1	[Jul16]	19/05/2016	31.91	
14	BO c1	[Jul16]	20/05/2016	31.27	
15	BO c1	[Jul16]	23/05/2016	31.15	
16	BO c1	[Jul16]	24/05/2016	30.89	

FX rates are imported similarly.

	A	B	C
1		EURUSD	USDMYR
2	30/06/2017	1.1426	4.2925
3	29/06/2017	1.1426	4.2935
4	28/06/2017	1.1378	4.2975
5	27/06/2017	1.1426	4.2925
6	26/06/2017	1.1182	4.2885
7	23/06/2017	1.1194	4.2880
8	22/06/2017	1.1153	4.2870
9	21/06/2017	1.1168	4.2875
10	20/06/2017	1.1135	4.2835
11	19/06/2017	1.1148	4.2925
12	16/06/2017	1.1198	4.2765
13	15/06/2017	1.1145	4.2675
14	14/06/2017	1.1218	4.2580
15	13/06/2017	1.1216	4.2925
16	12/06/2017	1.1203	4.2658
17	09/06/2017	1.1196	4.2650
18	08/06/2017	1.1214	4.2685
19	07/06/2017	1.1257	4.2605
20	06/06/2017	1.1278	4.2665